

# Hayden Michel

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## EDUCATION

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### The Ohio State University

Expected Graduation May 2027

BS in Electrical & Computer Engineering. Minor in Japanese

Columbus, OH

- GPA: 3.96
- Involved in Athletic Band, Frassati Mountaineers, Tau Beta Pi, Longboarding Club, OHI/O Makeathon

## WORK EXPERIENCE

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### Integrated Solutions For Systems

May 2026 - August 2026

Engineering intern

- Developed new plugins for PNTOS, an open-source system designed to build and fuse position, navigation, and timing sensor applications.
- Improved CI/CD pipelines to lower run time of 3 jobs by 96% using modern python tools.
- Developed an algorithm to calculate ionospheric delay of signals traveling from satellite to ground with C++.

### OSU Electrosience Laboratory

Feb 2026 - Present

Research Assistant - Wearable and Implantable Technologies Group

- Improving signal integrity of a wearable device using magnetic fields to detect heartbeats as a means to estimate cognitive workload.
- Integrating an IMU accelerometer and processing the combined signals of it and the heartbeat detector to adaptively filter out motion artifacts from breathing or moving using Matlab.
- Developing a phantom to model human heartbeat signals with coil antennas and electromagnetic principles.

### FirstEnergy

June 2025 - May 2026

Fiber Networks Intern

- Independently managed 3 fiber projects connecting power assets including planning routes, running weekly calls and connecting between different contractors.
- Automated tracking unplanned fiber outages in an Excel database using Microsoft power automate, saving 1-3 hours of manual data entry per week.

### OSU College of Optometry

August 2024 - Present

Research Assistant

- Used python machine learning and computer vision to analyze walking paths of blindfolded subjects.
- Used C#, Unity, and SteamVR to assist the College of Optometry research by making VR simulations using eye tracking capabilities in VR headset to test and analyze patient performance.
- Developed a full stack application that uses Javascript and Python to remotely control multiple Arduino devices over bluetooth.

### OSU Department of Electrical and Computer Engineering

January 2025 - May 2025

Teaching Assistant for ECE 2060 Intro to Digital Logic

- Guided 50+ students weekly through hands-on digital logic labs, enforcing core concepts in combinational logic, sequential circuits, and Finite State Machine (FSM) design.
- Mentored students in writing and debugging VHDL code using Intel Quartus Prime to program FPGA boards.

## SKILLS

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Java, C++, Python, C#, Matlab, Autodesk Inventor, Computer Vision, Arduino, CI/CD, Power Automate  
Antenna Design, Signal Processing, Semiconductor Devices, Hardware Integration, FPGA, VHDL  
Japanese Language (JLPT N2 certified. Lived in Japan for a year as an exchange student with a host family.)

## AWARDS/DISTINCTIONS

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Eagle Scout, May 2022 - Installed boardwalks at local hiking trails as an Eagle service project.  
Selected for the Fundamentals of Engineering Design Showcase for software design project final.